

OMOMIJOLAOLUWA AKINROGUNDE

BACKEND ENGINEER | FastAPI · Django · PostgreSQL · API Infrastructure

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PROFESSIONAL SUMMARY

Backend Engineer specialized in designing and scaling RESTful APIs with FastAPI, Django, and PostgreSQL. Track record of cutting application response time by 76% through query and architecture optimization, automating deployments with Docker and CI/CD, and hardening production systems against security risks. Comfortable owning a backend end-to-end — from schema design through deployment — and collaborating with frontend teams (React/Next.js) when needed.

EXPERIENCE

Backend Engineer — FUD'S

January 2026 - Present, Remote

- Architecting and building a high-performance FastAPI backend for a multi-vendor food delivery and e-commerce platform, serving tailorable meal subscriptions (weekly/monthly plans).
- Implemented recommendation intelligence to suggest customized balanced/low-carb options matching user diet profiles (e.g., weight loss).
- Developed a consolidated cart system allowing users to place single orders from multiple vendors with single-rider delivery and advanced scheduling.
- Built a subscription-based grocery engine supporting automated same-day deliveries at weekly, bi-weekly, or monthly intervals.
- Designed the "111 Feature" enabling users to pre-schedule breakfast, lunch, and dinner slots dynamically aligned with vendor closing hours and rider availability.

Full-Stack Developer & DevOps — Konasalti Tech School

August 2025 - October 2025, Remote

- Architected and deployed a FastAPI backend serving 100+ active users, with JWT-based authentication and a normalized PostgreSQL schema.
- Cut application load time from 6s to 1.4s (76% reduction) through backend query optimization and targeted frontend performance tuning.
- Built a CI/CD pipeline with GitHub Actions, automating deployments to Vercel and Render and removing manual release steps.
- Containerized services with Docker and configured an NGINX reverse proxy for secure, scalable production deployment.
- Implemented asynchronous task processing with Celery and Redis to handle background jobs and notifications without blocking request cycles.

Backend & Infrastructure Engineer (Contract) — Cloudtalk Networks Limited

June 2024 - January 2026, Lagos, Nigeria

- Resolved critical database inconsistencies and improved website response time, restoring system reliability for active users.
- Hardened application security by building server-level IP banning scripts that automatically block suspicious traffic.
- Reduced data breach risk through system updates, vulnerability mitigation, and access control improvements.
- Supported infrastructure troubleshooting and platform stability work alongside 1st–2nd line technical support duties.

PROJECTS

Ree-fond — Regulatory Compliance Workflow System — FastAPI, PostgreSQL, SQLAlchemy Docker, JWT, REST APIs

- Architected a multi-tenant backend using Clean Architecture (FastAPI, async SQLAlchemy, PostgreSQL), enforcing strict data isolation across organizations, taxpayers, filings, and refunds at the database query level.
- Built a rule-based compliance engine that scores taxpayers 0–100 and classifies risk into Low/Medium/High/Critical tiers, plus an end-to-end refund workflow (Initiated → Disbursed) with automated SLA tracking for regulatory deadlines.

- Secured the platform with JWT auth (Argon2 hashing) and Pydantic V2 validation, and implemented a tiered usage-quota system enforcing API/filing limits per subscription plan.

Backend Battle — API Performance Benchmarking Platform — *FastAPI, httpx, React Native (Expo), Async Processing*

- Built an API stress-testing platform that simulates concurrent load (up to 50 users, 200 requests) against registered endpoints and scores performance using a weighted latency/throughput/success-rate algorithm.
- Designed an async benchmarking engine using httpx and FastAPI BackgroundTasks to run simulations out-of-band without blocking API responses
- Built REST endpoints for submitting, retrieving, and ranking benchmark runs, with a real-time leaderboard supporting time-based filtering (today/week/month).
- Identified production-readiness gaps (in-memory store, lack of job queueing, no rate limiting) and documented a hardening path using PostgreSQL, Redis/Celery, and slowapi.

KANEC Impact — Blockchain Donation Transparency Platform — *FastAPI, PostgreSQL, Docker, Hedera SDK, Celery/Redis*

- Architected a backend integrating Hedera's distributed ledger SDK for on-chain donation recording, including automated per-user wallet creation, encrypted private key storage, escrow accounts, and P2P wallet transfers with full audit trails via mirror node integration.
- Developed AI-driven analytics (pandas, scikit-learn) surfacing donation trends and project performance metrics, powering a real-time predictive insights dashboard.
- Implemented JWT authentication, email verification/OTP flows (Brevo API), and async background processing (Celery/Redis) for non-blocking notification and verification jobs.

Outbound AI Calling Agent — Backend Engineer (Team Project) — *FastAPI, PostgreSQL, Node.js*

- Engineered the relational schema and RESTful APIs powering an AI outbound call automation platform, covering call sessions, user management, and event-triggered workflows.
- Integrated subscription and usage-based payment processing into the call automation pipeline, collaborating cross-functionally to keep the architecture scalable.

TECHNICAL SKILLS

Backend & APIs: FastAPI, Django, DRF, Flask, REST API Design

Databases: PostgreSQL, MySQL, Query Optimization & Tuning

Infrastructure & DevOps: Docker, NGINX, GitHub Actions, CI/CD, Celery, Redis

Cloud & Hosting: AWS (EC2), Cloudflare, DigitalOcean, Render, Vercel

Security & Compliance: JWT Auth, Stripe Payment Integration, Access Control, API Hardening, IP Filtering

Frontend (Collaboration): React, Next.js (performance optimization)

EDUCATION

Bachelor of Arts in Mass Communication — Babcock University